

Welfare measurements with heterogeneous agents.

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Motivation

THE MISSING INDEX

- The canonical infinite-horizon framework with heterogeneous consumers, standard in macroeconomics and finance, lacks a preference-based index that consistently quantifies the welfare impact of economic policies.
- The classic money-metric indices, equivalent and compensating variation, fail two consistency requirements in this setting: they are non-additive across sets of policies, and their predictions can depend on the assumed status quo or on the order in which alternatives are implemented.
- The paper delivers a positive result on when these failures become quantitatively negligible.

Money-metric welfare indices

DEFINITIONS

For consumer i with lifetime utility V_i , wealth w_i and a policy moving the environment from ϑ^0 to ϑ^1 , the equivalent and compensating variations solve

$$V_i(w_i + EV_i; \vartheta^0) = V_i(w_i; \vartheta^1), \quad V_i(w_i; \vartheta^0) = V_i(w_i - CV_i; \vartheta^1).$$

- Additivity would require the index of a bundle of policies to equal the sum of the indices of its elements; exact additivity fails for general preferences, and with it order-independence.
- The question is under what restrictions the money metric becomes approximately additive, so that policy rankings aggregate.

| Main result

NEAR-ADDITIVITY UNDER PATIENCE

- For arbitrary heterogeneous von Neumann–Morgenstern preferences sharing a common discount factor, the equivalent (or compensating) variation is nearly additive and aggregates effectively as long as consumers are patient.
- The approximation error of treating the index as additive vanishes as the discount factor approaches one, so for short-lived policies the money metric behaves as a consistent quantitative welfare index.
- Consequently, the index delivers consistent welfare predictions for the wide variety of short-lived policies studied in the macroeconomic and finance literature.

Structure of the argument

FROM EXAMPLE TO GENERAL FRAMEWORK

- A motivating example isolates the non-additivity of the money metric in the simplest heterogeneous-agent setting.
- The general framework is a small open economy with heterogeneous vNM consumers; the near-additivity result is established there.
- An approximation-in-practice section quantifies the error terms and shows how the index is operationalised in applied policy evaluation.

| Implications

FOR APPLIED POLICY EVALUATION

- Welfare statements in heterogeneous-agent models gain a consistent money-metric foundation for the empirically dominant case of short-lived policies evaluated by patient consumers.
- Policy bundles can be evaluated component-wise and summed, and rankings are invariant to the order of implementation up to the quantified approximation error.
- The result connects the applied macro practice of reporting consumption-equivalent welfare numbers to an explicit preference foundation.

Thank you.

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